Bhavani Badugu

ee24mtech02001@iith.ac.in | bhavanibadugu123@gmail.com | +91 9573292283 | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.26
Rajiv Gandhi University of Knowledge and Technology , Basar	Class XII (Pre University Course)	2018	8.89%
Z P High School, Ismailkhanpet, Medak	Class X (Secondary School Certificate)	2016	9.70%

SCHOLASTIC ACHIEVEMENTS

Secured **1st place** at the **Divisional level in SSC**, Leading admission to the prestigious RGUKT Basar (IIIT) six year integration course.

Awarded **2nd place, State-level** Elocution Competition, 2016

POSITIONS OF RESPONSIBILITY

Teaching assistant for the Course (Autonomous Navigation) Prof. P Rajalakshmi

SKILLS

Programming Languages: C, Embedded C, Python, SystemVerilog, Verilog

Chip Design: Core Digital Design, RTL Design, SOC, RISC-V

Verification: UVM, Verilog

Protocols: CAN, I2C, UART

VLSI Tools: Xilinx Vivado, GTKWave, ModelSim

RELEVANT COURSES

Coursework: Digital IC Design, FPGA LAB, Verilog, Real-Time Embedded System

PROJECTS

32-bit RISC-V Processor Design & Verification

- Designed a 32-bit single-cycle RISC-V CPU in SystemVerilog supporting R, I, S, B, U, and J-types instruction sets.
- Extended the single-cycle architecture to a five-stage pipeline with forwarding and hazard-detection logic, enhancing throughput through instruction-level overlap and higher clock frequency.
- Verified across 35 automated tests with a 100% pass rate.

UART IP Design & Verification

- Designed a configurable UART IP in SystemVerilog supporting programmable data bits (5-8), parity, stop bits, 16-deep TX/RX FIFOs, and interrupt generation.
- Implemented UVM-based verification environment with scoreboard, functional coverage, and error injection (parity, framing, overrun).
- Achieved high code and functional coverage across multiple configurations and baud rates.

RTL Implementation of I2C Protocol for Master-Slave Communication with Register Bank Memory:

- Designed RTL for master and slave components to perform read and write operations, with the slave featuring a register bank memory to retrieve data written during the write cycle in the subsequent read cycle, thereby achieving reliable data transfer.

SDRAM Controller Design & Verification in Verilog

- Designed a modular 3rd-generation SDRAM controller in Verilog with Initialization, Auto/Self-Refresh, Mode Register, and burst Write/Read support using robust FSMs and strict timing compliance.
- Integrated modules under a top-level arbiter and fully verified against the Micron SDRAM model in Xilinx Vivado.

EXPERIENCE

Research Fellow | **TiHAN, IIT Hyderabad July 2022 - December 2023**

- Worked on the **Drive-by-Wire** system for a 14-seater autonomous campus shuttle, evaluating and integrating throttle, braking, and steering control using **dSPACE MicroAutoBox control and Configuration Desk.**
- Collaborated on end-to-end system evaluation, ensuring reliable signal acquisition and actuation and Motors for autonomous control

EXTRACURRICULAR

Table Tennis Jogging Running